Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 1

Semester Test 1

Module EBN122

Elektrisiteit en Elektronika

19 Augustus 2013

Module EBN122

Electricity and Electronics

19 August 2013

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*			
Maksimum punte: *Maximum marks:*	45	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 minute *90 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		15	

BELANGRIK- *IMPORTANT*

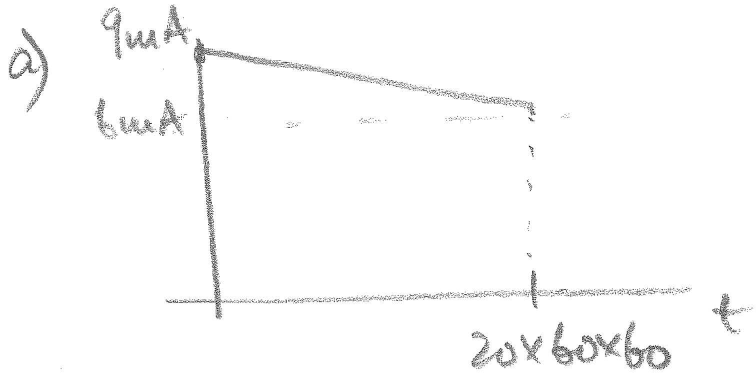
1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	
	1 J Joubert
	2 JD le Roux

VRAAG/QUESTION 1 [9]

Vraag 1.1 [4]	Question 1.1 [4]
a) 'n Volgelaaide 9V battery word oor 'n 1kΩ weerstand verbind. Die aanvanklike spanning oor die battery is 9V, en na 20 ure het die spanning verminder na slegs 6V. Aanvaar dat die afname in spanning lineêr met tyd gebeur het. Bereken die hoeveelheid lading wat gedurende die 20 ure deur die weerstand beweeg het.	a) A fully charged 9V battery is connected across a 1kΩ resistor. The initial voltage across the battery is 9V, and after 20 hours the voltage has reduced to only 6V. Assume that the decrease happened linearly with time. Calculate the charge that moved through the resistor during the 20 hours.
b) As die energiekoste 50 sent/kWh is, hoeveel 60W ligte kan jy laat brand vir 8 ure/dag as jou begroting vir beligting R250/maand is. Aanvaar dat 'n gemiddelde maand 30 dae lank is.	b) If the cost of energy is 50 cents/kWh, how many 60W lightbulbs can you have on for 8 hours/day if your budget for lighting is R250/month. Assume that an average month has 30 days.

a)



$$q = 6\times10^{-3}\times 20\times60\times60 + \tfrac{1}{2}\times 3\times10^{-3}\times 20\times60\times60$$

$$= 432 + 108 = \underline{540\ C}$$

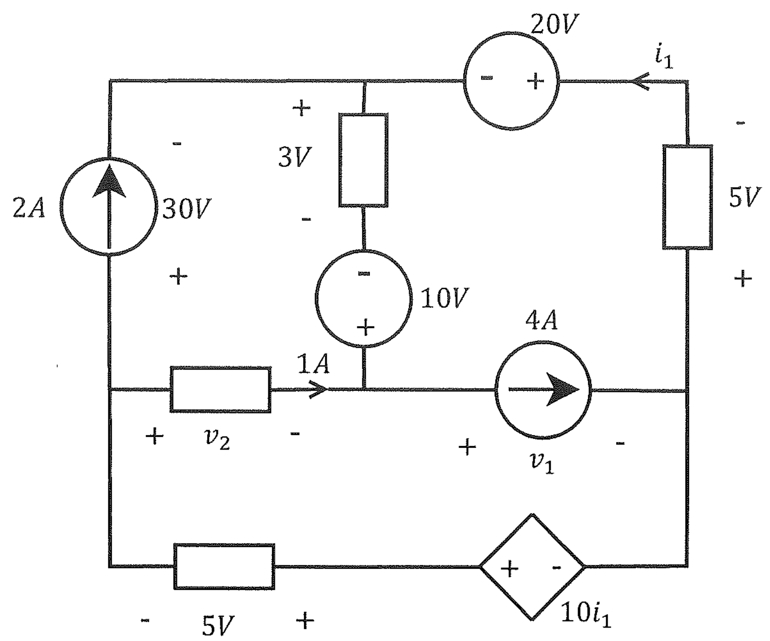
b) Energie (1 gloeilamp) (1 bulb) $= 0.06\ kW \times 8 \times 30$

$= 14.4\ kWh$

aantal gloeilampe (number of bulbs) $= \dfrac{500\ kWh}{14.4\ kWh}$

$= \underline{34 \text{ gloeilampe}}$

Vraag 1.2 [3] Bepaal die spannings v_1, v_2 en die stroom i_1.	Question 1.2 [3] Determine the voltages v_1, v_2 and the current i_1.



$i_1 = \underline{1A}$ ✓

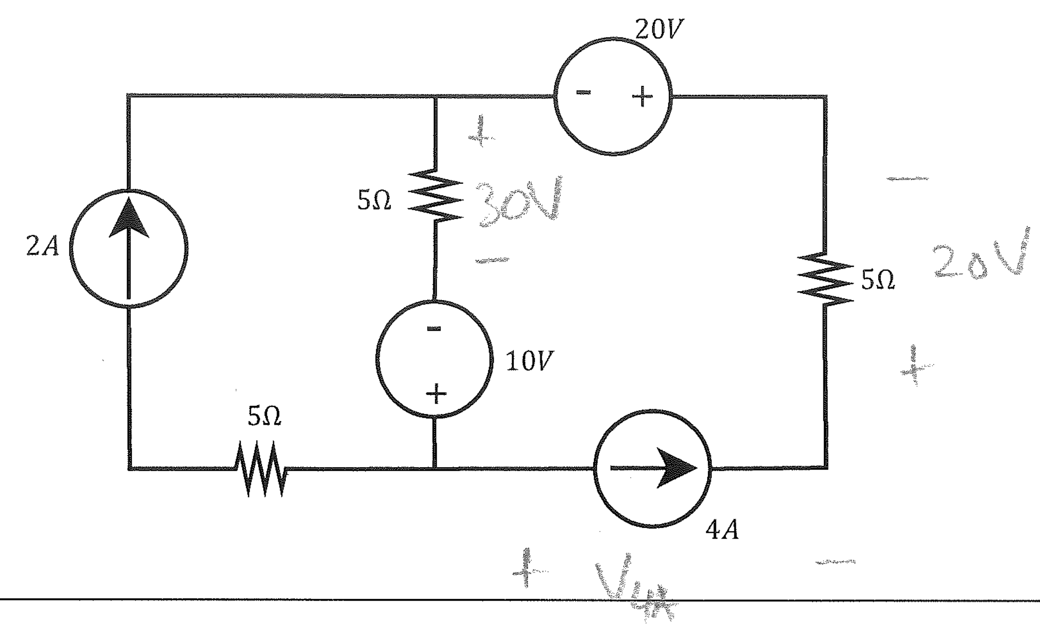
$V_1 + 5 + 20 + 3 - 10 = 0$

$V_1 = \underline{-18V}$ ✓

$V_2 + 10 - 3 - 30 = 0$

$V_2 = \underline{23V}$

Vraag 1.3 [2]	Question 1.3 [2]
Bereken die drywings ge-assosieer met die *4A* stroombron en die *20V* spanningsbron.	Calculate the power associated with the *4A* current source and the *20V* voltage source.



$V_{4A} + 20 + 20 + 30 - 10 = 0$

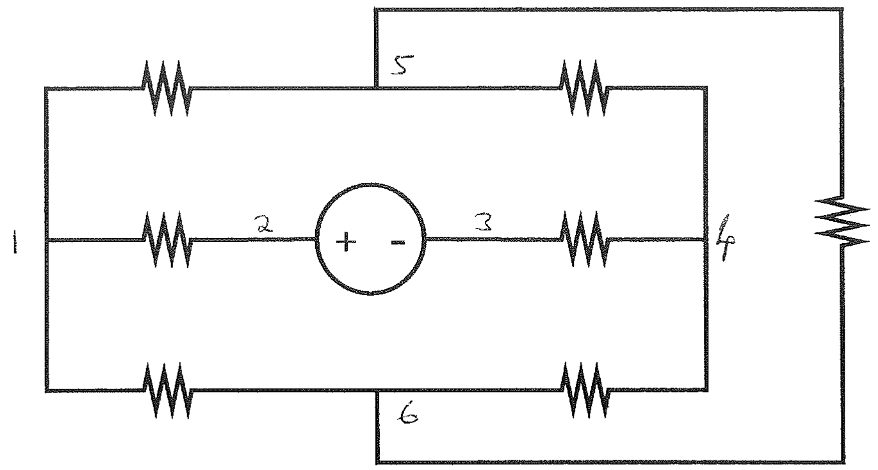
$V_{4A} = -60V$

$P_{20V} = 20 \times 4 = 80W$ ✓

$P_{4A} = 4 \times (-60) = -240W$ ✓

VRAAG/QUESTION 2 [22]

Vraag 2.1 [3] Bepaal die aantal takke, nodusse en onafhanklike lusse vir die volgende kringbaan.	**Question 2.1 [3]** Determine the number of branches, nodes and independant loops for the circuit below.

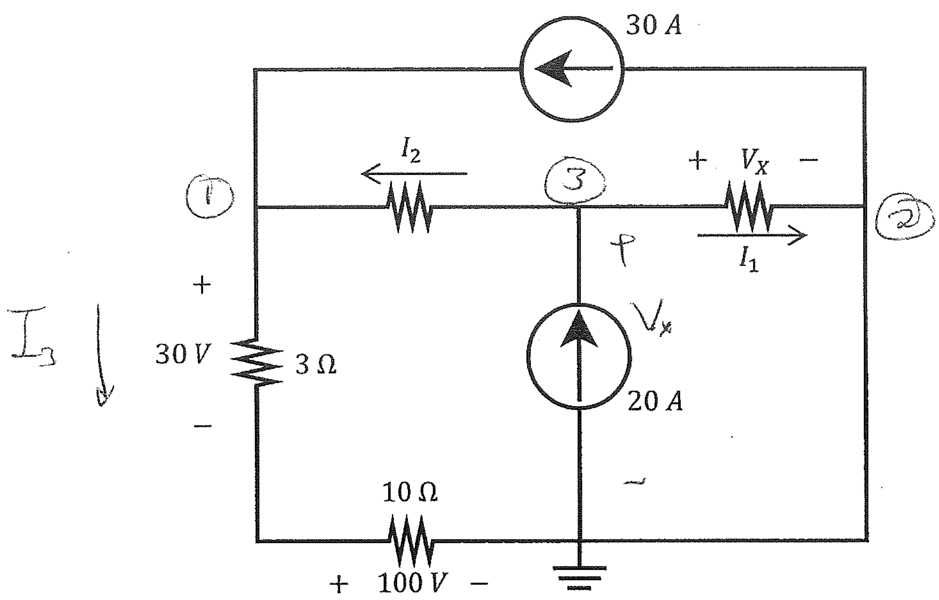


$b = 8$ ✓

$n = 6$ ✓ $\quad b = l + n - 1.$

$l = 3$ ✓

Vraag 2.2 [3] Indien $V_X = 10$ V, bepaal I_1, I_2 en die drywing geassiosieer met die 20 A stroombron.	**Question 2.2 [3]** If $V_X = 10$ V, determine I_1, I_2, and the power associated with the 20 A current source.



$* \quad P_{20A} = -(V_x) \times 20A.$

$= -200W$ ✓

$* \quad I_3 = \frac{30}{3} = 10A \quad$ or $\quad I_3 = \frac{100}{10} = 10A.$

KCL: * (Node 1) $\quad I_2 + 30 = I_3.$

$I_2 = I_3 - 30 = -20A$ ✓

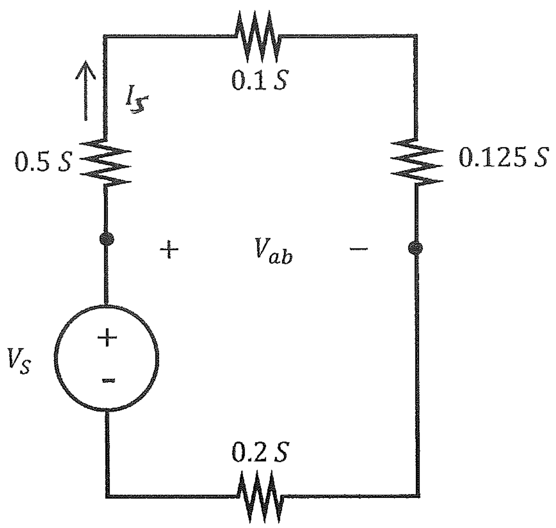
KCL * (Node 2) $\quad I_1 + I_3 = 30 + 20$

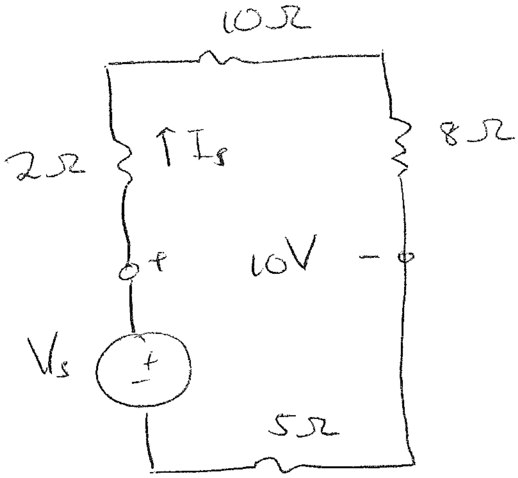
$I_1 = 50 - 10 = 40A.$ ✓

Check: KCL: (Node 3) $\quad 20 = I_1 + I_2.$

$I_1 = 20 - (-20) = 40A$

Vraag 2.3 [2]	Question 2.3 [2]
Bepaal V_S en I_S indien $V_{ab} = 10$ V.	Determine V_S and I_S if $V_{ab} = 10$ V.





KVL: $2I_s + 10I_s + 8I_s - 10 = 0$

$20I_s = 10$

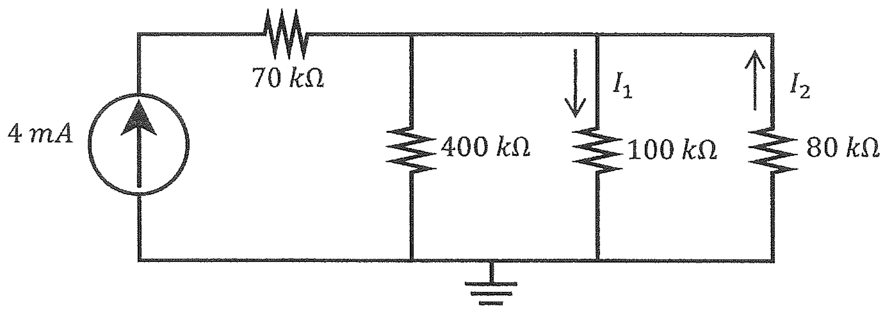
$I_s = 0{,}5\text{A}.$ ✓

KVL: $-V_s + 10 + 5I_s = 0$

$V_s = 10 + 5 \times 0{,}5$

$= 12{,}5\text{V}$ ✓

Vraag 2.4 [2] Bepaal I_1 en I_2.	**Question 2.4 [2]** Determine I_1 and I_2.



$$R_{11} = \left(\frac{1}{400} + \frac{1}{100} + \frac{1}{80}\right)^{-1} \times 10^3$$

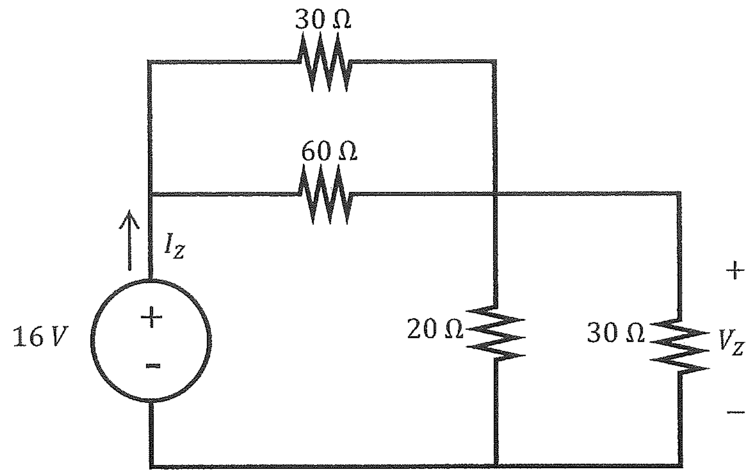
$$= 40k\Omega.$$

$$V_{11} = 4mA \times 40k\Omega = 160V.$$

$$I_1 = \frac{160V}{100k\Omega} = 1,6mA. \checkmark$$

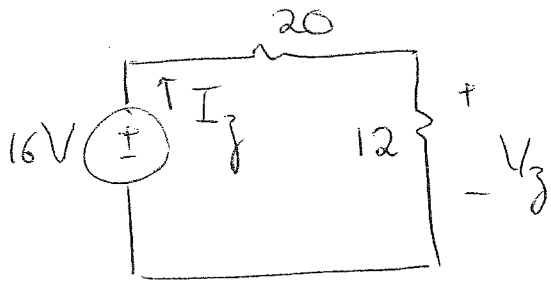
$$I_2 = \frac{-160V}{80k\Omega} = -2mA \checkmark$$

Vraag 2.5 [2] Bepaal I_Z en V_Z.	Question 2.5 [2] Determine I_Z and V_Z.



$$R_{\parallel_1} = 30 \parallel 60 = \frac{30 \times 60}{90} = 20\Omega$$

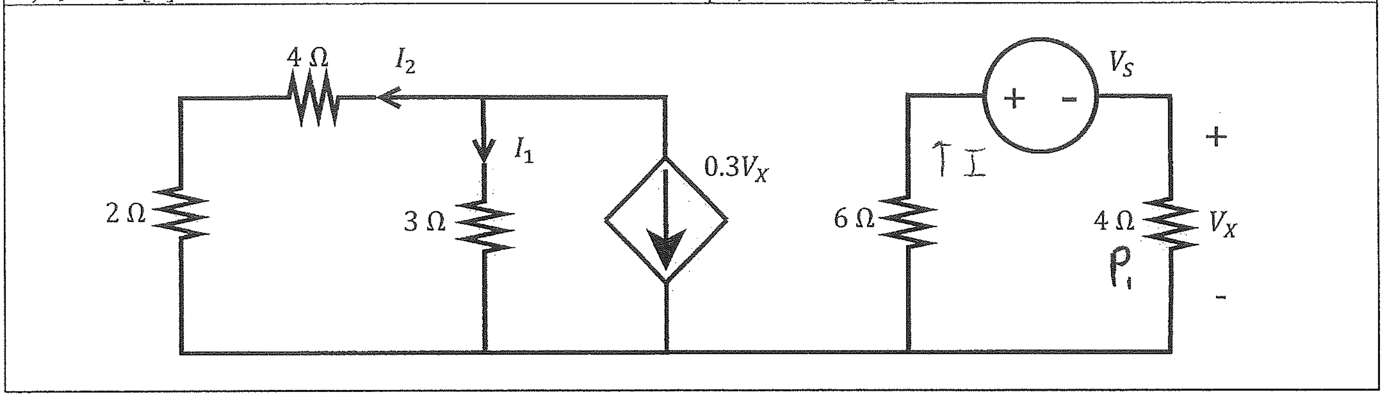
$$R_{\parallel_2} = 20 \parallel 30 = \frac{30 \times 20}{50} = 12\Omega.$$



$$I_z = \frac{16}{20+12} = 0{,}5A.$$

$$V_z = \frac{16 \times 12}{20+12} = 6V$$

Vraag 2.6 [4]	Question 2.6 [4]
$V_X = 20$ V. Bepaal: a) V_S en die drywing P_1. [2] b) I_1 en I_2. [2]	$V_X = 20$ V. Determine: a) V_S and the power P_1. [2] b) I_1 and I_2. [2]



a) $P_1 = \frac{20^2}{4} = 100\,W$

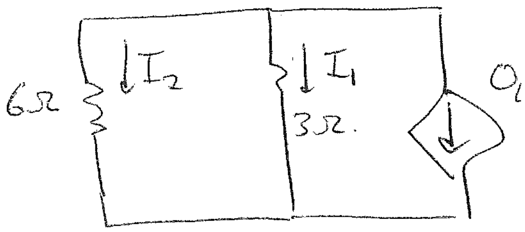
b) $+V_s + I \times 4 + 6 \times I = 0$

$V_s = -10I$

but $I = \frac{20}{4} = 5A$

$\therefore V_s = -50V$

b)



$0{,}3 \times 20 = 6A$

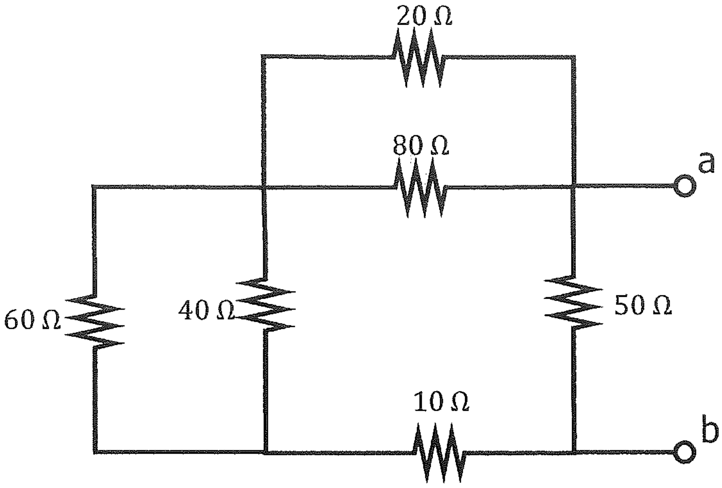
$I_1 = \frac{-6 \times 6}{6+3} = -4A$

$I_2 = \frac{-6 \times 3}{6+3} = -2A$

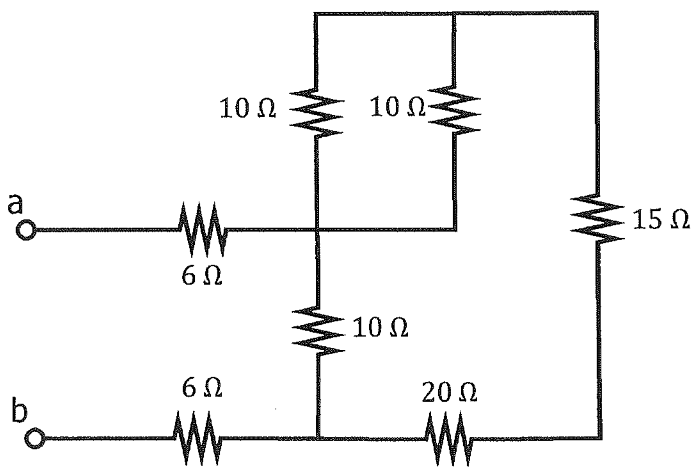
(Current division)

Vraag 2.7 [6]	Question 2.7 [6]
Bepaal die ekwivalente weerstand R_{ab} vir elkeen van die drie onderstaande resistornetwerke. (Die derde netwerk word getoon op die volgende bladsy.)	Determine the equivalent resistance R_{ab} for each of the three resistor networks shown below. (The third network is shown on the next page.

Kringbaan / *Circuit* 1



Kringbaan / *Circuit* 2



① $R_{ab} = 50 \parallel (10 + 60\parallel 40 + 20 \parallel 80)$

$= 50 \parallel (10 + 24 + 16)$

$= 50 \parallel 50 = 25\,\Omega$

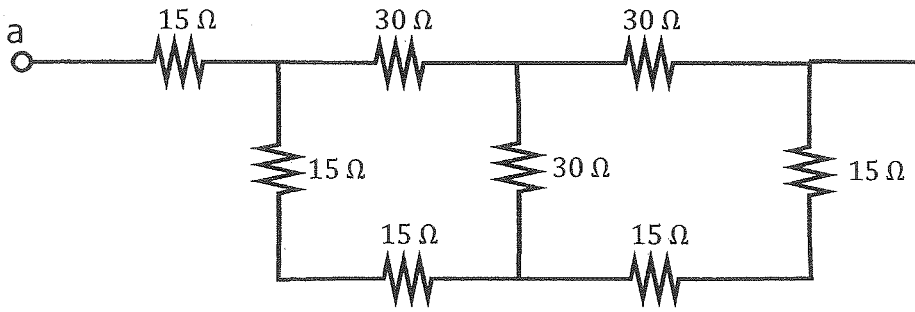
② $R_{ab} = 6 + 6 + 10 \parallel (20 + 15 + 10\parallel 10)$

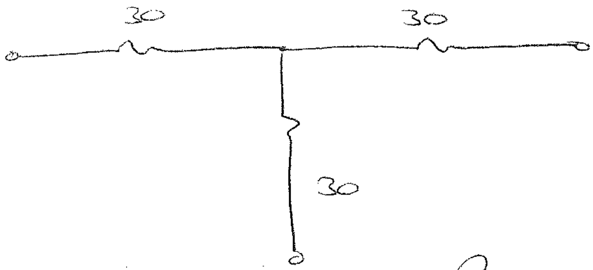
$= 12 + 10 \parallel (35 + 5)$

$= 12 + 10 \parallel 40$

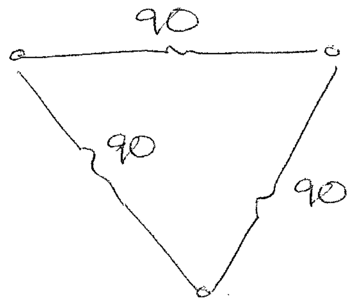
$= 12 + 8 = 20\,\Omega$

Kringbaan / *Circuit* 3

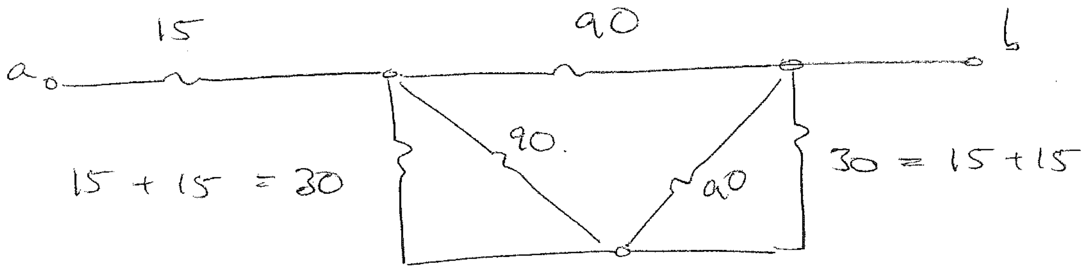




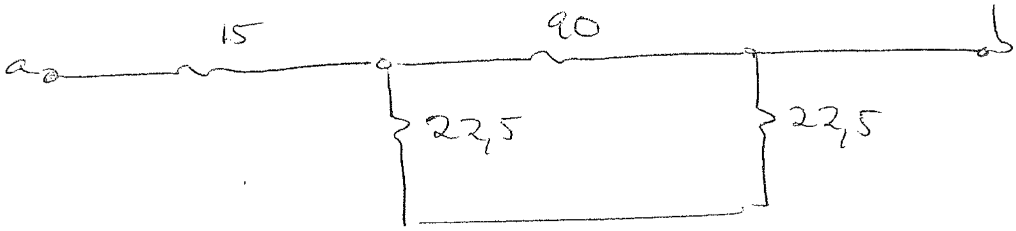
$\Rightarrow$



$$R_\Delta = 3R_Y.$$



$$R_n = 90 \| 30 = 22{,}5\,\Omega.$$

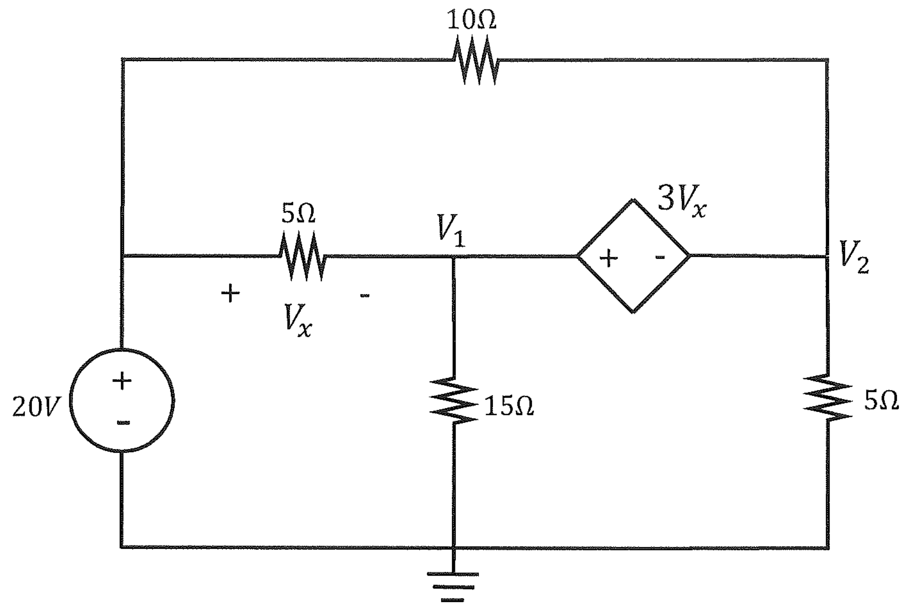


$$R_{ab} = 15 + 90 \| (22{,}5 + 22{,}5)$$

$$= 15 + 90 \| 45 = 15 + 30 = \underline{45\,\Omega}$$

VRAAG/QUESTION 3 [14]

Vraag 3.1 [4]	Question 3.1 [4]
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: b −60 $aV_1 + V_2 = b$ (supernode) $cV_1 + dV_2 = 180$ (KCL)	Use nodal analysis to set up two equations of the following form: b −60 $aV_1 + V_2 = b$ (supernode) $cV_1 + dV_2 = 180$ (KCL)



supernode:

$$V_2 + 3V_x = V_1$$

$$V_2 + 3(20 - V_1) = V_1$$

$$V_2 + 60 - 3V_1 = V_1$$

$$-4V_1 + V_2 = -60 \quad ✓✓$$

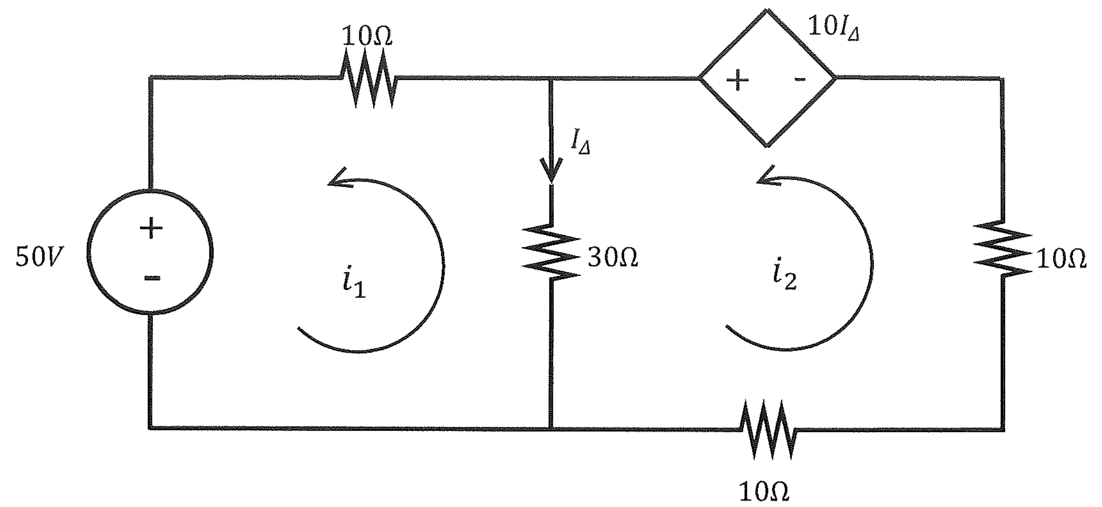
KCL

$$\frac{V_1}{15} + \frac{V_1 - 20}{5} + \frac{V_2}{5} + \frac{V_2 - 20}{10} = 0$$

$\times 30$: $2V_1 + 6V_1 - 120 + 6V_2 + 3V_2 - 60 = 0$

$$8V_1 + 9V_2 = +180 \quad ✓✓$$

Vraag 3.2 [4]	**Question 3.2 [4]**
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel: $$ei_1 + fi_2 = -50 \quad \text{(lus 1)}$$ $$-20i_1 + gi_2 = h \quad \text{(lus 2)}$$	Use mesh analysis to set up two equations of the following form: $$ei_1 + fi_2 = -50 \quad \text{(loop 1)}$$ $$-20i_1 + gi_2 = h \quad \text{(loop 2)}$$



$$50 + 30(i_1 - i_2) + 10i_1 = 0$$

$$40i_1 - 30i_2 = -50 \quad ✓✓$$

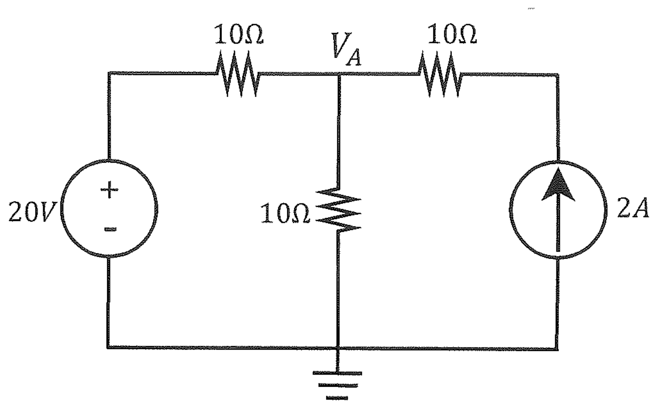
$$-10I_\Delta + 30(i_2 - i_1) + 20i_2 = 0$$

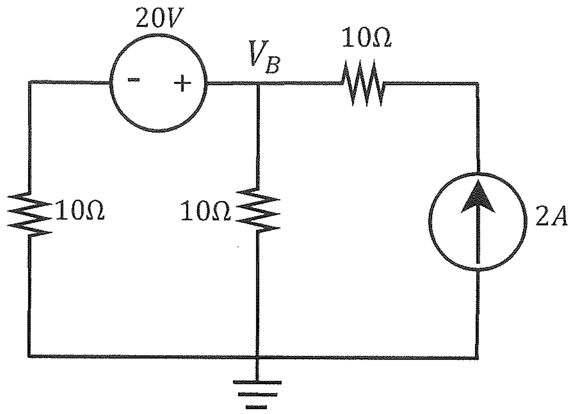
$$-10(i_2 - i_1) + 30(i_2 - i_1) + 20i_2 = 0$$

$$20(i_2 - i_1) + 20i_2 = 0$$

$$-20i_1 + 40i_2 = 0 \quad ✓✓$$

Vraag 3.3 [2] Gebruik node-analise om node-spannings V_A en V_B te bereken.	**Question 3.3 [2]** Use nodal analysis to calculate node voltages V_A and V_B.





$$\frac{V_a - 20}{10} + \frac{V_A}{10} \cancel{4} - 2 = 0$$

$$V_a - 20 + V_a - 20 = 0$$

$$V_a = \frac{40}{2} = 20V$$

$$V_B = V_A$$

Vraag 3.4 [4]	**Question 3.4 [4]**
a) Bepaal V_2. $\begin{bmatrix} 4 & -3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$ b) Bepaal i_2. $\begin{bmatrix} 1 & -1 & 0 \\ 3 & 2 & 2 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$	a) Determine V_2. $\begin{bmatrix} 4 & -3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$ b) Determine i_2. $\begin{bmatrix} 1 & -1 & 0 \\ 3 & 2 & 2 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

a)

$$\Delta = \begin{vmatrix} 4 & -3 \\ -2 & 4 \end{vmatrix} = 16 - 6 = 10$$

$$\Delta_2 = \begin{vmatrix} 4 & 0 \\ -2 & 10 \end{vmatrix} = 40 - 0 = 40$$

$$V_2 = \frac{40}{10} = 4\,V$$

b)

$$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ 3 & 2 & 2 \\ -1 & 0 & 1 \end{vmatrix} \begin{matrix} 1 & -1 & 0 \\ 3 & 2 & 2 \end{matrix} = 2 + 0 + 2 - (0) - (0) - (-3)$$

$$= 7$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & 0 \\ 3 & 2 & 2 \\ -1 & 1 & 1 \end{vmatrix} \begin{matrix} 1 & 1 & 0 \\ 3 & 2 & 2 \end{matrix} = 2 + 0 - 2 - (0) - (2) - (3)$$

$$= -5$$

$$i_2 = \frac{\Delta_2}{\Delta} = \frac{-5}{7} = -0.714$$